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CO1 AT-03

Descriptive Written Test

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1. Types of Intelligent Agents in Smart Home Automation

An intelligent agent perceives its environment through sensors and takes actions to achieve desired results. In a smart home, it controls lighting, temperature, and security.

Types of Intelligent agents

1. Simple Reflex Agent: Acts based on the current input using condition-action rules. For example, it switches on lights when motion is detected. It is fast but cannot remember past events.

2. Model-Based Agent: Maintains an internal model of the environment and uses past information. For example, it tracks room occupancy and temperature to control air conditioning. It works better when the environment changes.

3. Goal-Based Agent: Selects actions that achieve a comfortable environment while a specific goal. For example it adjusts lighting and temperature to maintain a comfortable environment while securing the house when the user leaves.

4. Utility-Based Agent: Chooses actions that maximize overall benefit by considering several factors. For example, it balances comfort, electricity consumption and security to select the best settings.

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comparison and applications

Agent type	Main advantage	Limitation
Simple reflex	Fast response	No memory
Model-based	Uses environmental information	Requires an internal model
Goal-based	Achieves specific goals	May require more computation
Utility-based	Balances multiple factors	Utility design can be complex

A hybrid model-based and utility-based agent is most effective. The model-based component tracks environmental conditions, while the utility-based component balances comfort, energy efficiency, and security. Learning user preferences further improves its decisions

2. Uninformed Search Strategies for Solving a Maze

Uninformed search algorithms explore possible paths without prior knowledge of the maze. A robot uses these strategies to find the exit by exploring available routes.

1. Uniform Cost Search (UCS):

UCS expands the path with the lowest total cost first. It uses a priority queue to select the least-cost path

* Advantages: Finds the optimal path when step costs are positive; handles different movement costs.

* Disadvantages: Can consume considerable time and memory because it stores many possible paths.
* Application: Suitable when different maze paths have different movement costs.

2. Iterative Deepening Search (IDS):

IDS repeatedly performs depth-limited search, increas the depth limit until the exit is found.

* Advantages: Uses less memory than UCS and ca find a shallow solution.
* Disadvantages: Repeats searches at earlier dep causing extra work.
* Application: Suitable for large mazes with eq step costs and unknown solution depth.

Complexity comparison:

Factor	UCS	IDS
Time complexity	$O(b^{1+\lfloor C^*/\epsilon \rfloor})$	$O(b^d)$
Space complexity	Exponential	$O(bd)$
Optimality	Yes, with positive costs	Yes, with equal step costs
Main benefits	Minimum-cost path	Low memory usage.

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Here, b is the branching factor, d is the solution depth, C^* is the optimal path cost, and ϵ is the minimum step cost.

Role of Intelligent agents:

* Simple reflex agent: Selects actions using current sensor inputs but may get stuck.
* Model-based agent: Maintains a map of explored locations and remembers visited paths.
* Goal-based agent: Selects actions to reach the exit.

A model-based, goal-based hybrid agent using IDS is suitable when maze movements have equal costs and memory is limited. If movement costs differ and finding the minimum-cost path is essential, UCS is the better choice. The agent's internal map and goal-directed decisions improve navigation and reduce repeated exploration.

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